

Engineering Mathematics Study Notes: Limit Comparison Test & P-Series Test

1. Concept Identification

The notes introduce a powerful, mechanical strategy for determining the convergence or divergence of an infinite series containing algebraic fractions:

- **The Limit Comparison Test (LCT):** A test used when an infinite series $\sum u_n$ is too complex to evaluate directly. We compare its behavior against a simpler, known helper series $\sum v_n$.
 - **The P-Series Test:** A highly standardized benchmark test used exclusively to instantly know whether the chosen helper series $\sum v_n$ converges or diverges based on the exponent value.
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2. Mathematical Representations

The Limit Comparison Test Condition

Let $\sum u_n$ be your given positive term series, and let $\sum v_n$ be your constructed helper series. Compute the limit of their ratio:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = L$$

The test is valid and operational if and only if L satisfies:

$$0 < L < \infty \quad (\text{where } L \text{ is a Finite, Non-Zero constant})$$

Under this condition, both series possess identical convergence behavior (they either converge together or diverge together).

The P-Series Test Formula

A standard p-series is written in the form:

$$\sum v_n = \sum \frac{1}{n^p}$$

The convergence or divergence behavior is determined mechanically by inspecting the value of the exponent p :

$$\text{If } p > 1 \implies \text{The series is Convergent}$$

$$\text{If } p \leq 1 \implies \text{The series is Divergent}$$

3. The Algorithm (Procedural Recipe)

To solve an infinite algebraic series using the Limit Comparison Test framework, follow this exact step-by-step recipe:

Step 1: Extract the General Term (u_n)

Identify the algebraic formula representing the inside of the summation sign.

$$u_n = \text{Given Expression}$$

Step 2: Construct the Helper Term (v_n)

Look at the formula for u_n . Find the highest power of n in the numerator and the highest power of n in the denominator. Define v_n as:

$$v_n = \frac{\text{Highest Power of } n \text{ in Numerator}}{\text{Highest Power of } n \text{ in Denominator}} = \frac{1}{n^p}$$

Step 3: Set up and Simplify the Ratio Fraction $\frac{u_n}{v_n}$

Multiply u_n by the reciprocal of v_n to cleanly cancel out the dominant n terms before passing the limit:

$$\frac{u_n}{v_n} = u_n \times \frac{1}{v_n}$$

Step 4: Evaluate the Limit as $n \rightarrow \infty$

Factor out the remaining dominant powers of n within the parentheses, cancel them, and apply the infinite limit substitution to collapse all remaining fractional n elements into zero:

$$L = \lim_{n \rightarrow \infty} \frac{u_n}{v_n}$$

Step 5: Run the P-Series Decision on v_n

Check the value of p from Step 2 against the p-series threshold rule. State whether $\sum v_n$ converges or diverges.

Step 6: Conclude the Target Behavior

Because L is a finite, non-zero number, declare that your target series $\sum u_n$ copies the exact conclusion found in Step 5.

4. Operational Mechanics

Mechanical Example 1: Creating a Helper Series (v_n)

The board shows examples of extracting the highest powers to form v_n . Let's trace how the teacher does it mechanically:

Case A

$$u_n = \frac{n^5}{n^7}$$

The highest power on top is n^5 . The highest power on bottom is n^7 .

$$v_n = \frac{n^5}{n^7} = \frac{1}{n^{7-5}} = \frac{1}{n^2}$$

Case B

$$u_n = \frac{n^2}{n^5}$$

The highest power on top is n^2 . The highest power on bottom is n^5 .

$$v_n = \frac{n^2}{n^5} = \frac{1}{n^{5-2}} = \frac{1}{n^3}$$

Mechanical Example 2: Factoring out Radicals

Let's look at the mechanical steps on the board for extracting terms trapped inside square roots:

$$u_n = \frac{1}{\sqrt{n+1} + \sqrt{n-1}}$$

Pull out n inside each radical sign:

$$u_n = \frac{1}{\sqrt{n(1+\frac{1}{n})} + \sqrt{n(1-\frac{1}{n})}}$$

Distribute the square root power onto both components:

$$u_n = \frac{1}{\sqrt{n}\sqrt{1+\frac{1}{n}} + \sqrt{n}\sqrt{1-\frac{1}{n}}}$$

Factor out the common $\sqrt{n}$ from the entire denominator:

$$u_n = \frac{1}{\sqrt{n}\left(\sqrt{1+\frac{1}{n}} + \sqrt{1-\frac{1}{n}}\right)}$$

Now, isolate the pure n fraction outside the parentheses to find your helper series v_n :

$$v_n = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$$

Mechanical Example 3: Full Solved Board Problem

Let's trace through the complete problem solved across the final board images:

$$\text{Test for convergence of the series: } \sum_{n=0}^{\infty} \frac{2n^3 + 5}{4n^5 + 1}$$

Step 1: State the general term

$$u_n = \frac{2n^3 + 5}{4n^5 + 1}$$

Step 2: Construct the helper term

The highest power on top is n^3 . The highest power on bottom is n^5 .

$$v_n = \frac{n^3}{n^5} = \frac{1}{n^2}$$

Step 3 & 4: Set up the limit of the ratio

Multiply u_n by the flip of v_n :

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \left(\frac{2n^3 + 5}{4n^5 + 1} \times \frac{n^2}{1} \right)$$

Distribute the n^2 directly into the top terms:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2n^5 + 5n^2}{4n^5 + 1}$$

Forcefully factor out the highest power (n^5) from the top and bottom expressions:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{n^5 \left(2 + \frac{5}{n^3}\right)}{n^5 \left(4 + \frac{1}{n^5}\right)}$$

Cancel the matching outer n^5 variables:

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{2 + \frac{5}{n^3}}{4 + \frac{1}{n^5}}$$

Substitute the infinite limit (∞) values:

$$L = \frac{2 + 0}{4 + 0} = \frac{2}{4} = \frac{1}{2}$$

Verify the condition: $\frac{1}{2}$ is a finite, non-zero constant. The test is completely valid.

Step 5: Apply the P-Series Test on our helper

Look back at our helper series formula:

$$\sum v_n = \sum \frac{1}{n^2}$$

Identify the exponent:

$$p = 2$$

Evaluate the condition:

$$2 > 1 \implies \sum v_n \text{ is Convergent}$$

Step 6: Conclude the final result

By the Limit Comparison Test, since $\sum v_n$ converges, the original target series also converges.

Conclusion: The sequence/series is convergent.